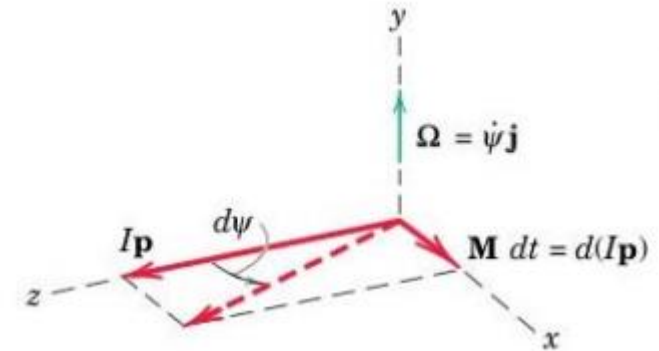
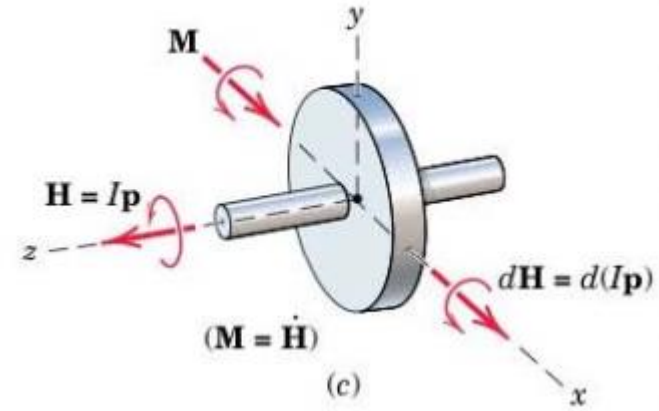
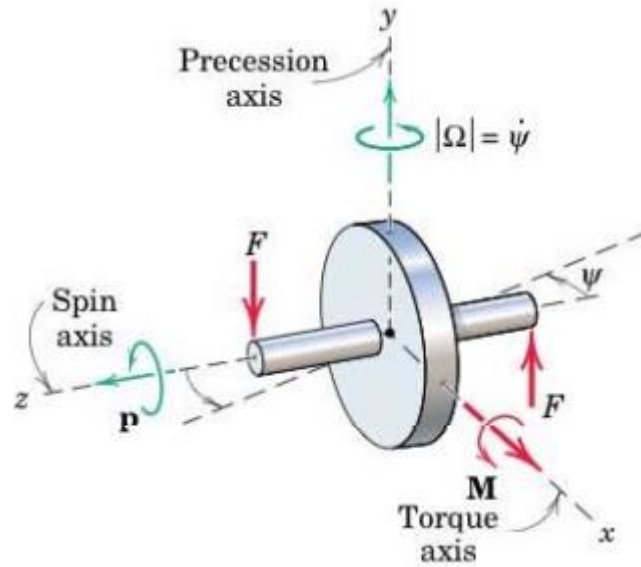


Gyroscopic motion

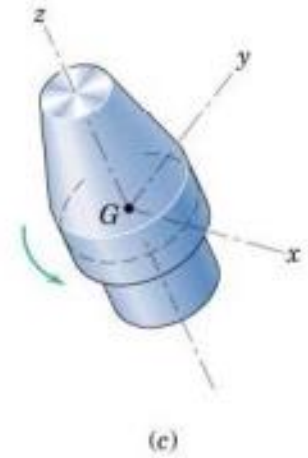
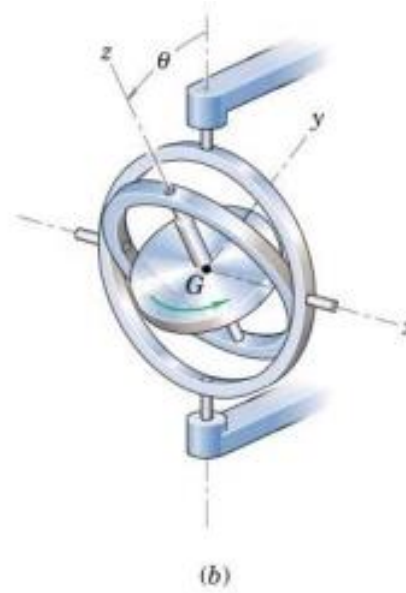
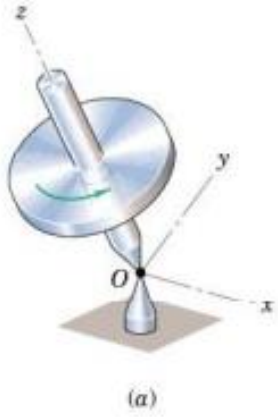
Dr. Atanu Banerjee

Gyroscopic motion

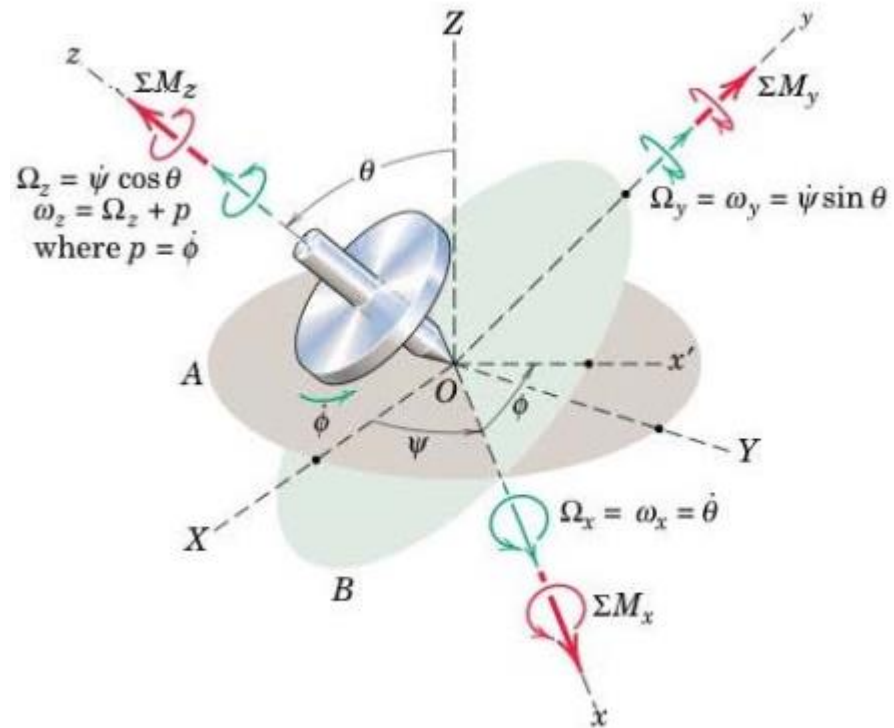


$$M = I \frac{d\psi}{dt} p$$

Gyroscopic motion



Governing equation of motion



$$\begin{aligned}\Omega_x &= \dot{\theta} & \omega_x &= \dot{\theta} \\ \Omega_y &= \dot{\psi} \sin \theta & \omega_y &= \dot{\psi} \sin \theta \\ \Omega_z &= \dot{\psi} \cos \theta & \omega_z &= \dot{\psi} \cos \theta + p\end{aligned}$$

$$\begin{aligned}H_x &= I_{xx}\omega_x = I_0\dot{\theta} \\ H_y &= I_{yy}\omega_y = I_0\dot{\psi} \sin \theta \\ H_z &= I_{zz}\omega_z = I(\dot{\psi} \cos \theta + p)\end{aligned}$$

$$\begin{aligned}\Sigma M_x &= I_0(\ddot{\theta} - \dot{\psi}^2 \sin \theta \cos \theta) + I\dot{\psi}(\dot{\psi} \cos \theta + p) \sin \theta \\ \Sigma M_y &= I_0(\ddot{\psi} \sin \theta + 2\dot{\psi}\dot{\theta} \cos \theta) - I\dot{\theta}(\dot{\psi} \cos \theta + p) \\ \Sigma M_z &= I \frac{d}{dt} (\dot{\psi} \cos \theta + p)\end{aligned}$$

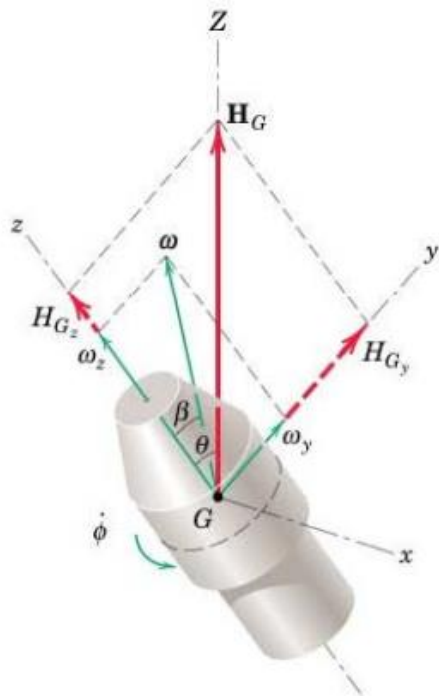
Steady state precession

$$\begin{aligned}\Sigma M_x &= I_0(\ddot{\theta} - \dot{\psi}^2 \sin \theta \cos \theta) + I\dot{\psi}(\dot{\psi} \cos \theta + p) \sin \theta \\ \Sigma M_y &= I_0(\ddot{\psi} \sin \theta + 2\dot{\psi}\dot{\theta} \cos \theta) - I\dot{\theta}(\dot{\psi} \cos \theta + p) \\ \Sigma M_z &= I \frac{d}{dt}(\dot{\psi} \cos \theta + p)\end{aligned}$$

$$\begin{aligned}\dot{\psi} &= \text{constant}, & \ddot{\psi} &= 0 \\ \theta &= \text{constant}, & \dot{\theta} = \ddot{\theta} &= 0 \\ p &= \text{constant}, & \dot{p} &= 0\end{aligned}$$

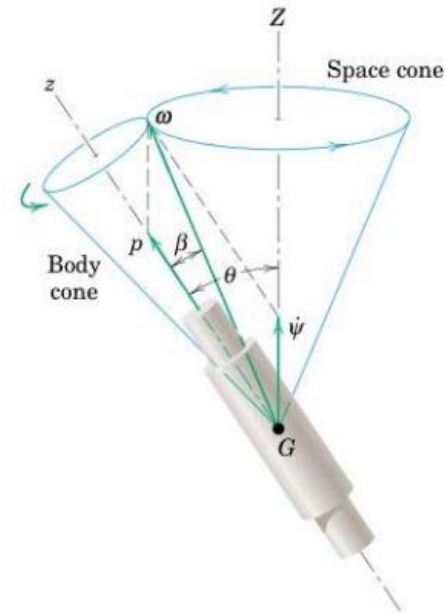
$$\begin{aligned}\Sigma M_x &= \dot{\psi} \sin \theta [I(\dot{\psi} \cos \theta + p) - I_0 \dot{\psi} \cos \theta] \\ \Sigma M_y &= 0 \\ \Sigma M_z &= 0\end{aligned}$$

Steady precession with zero moment

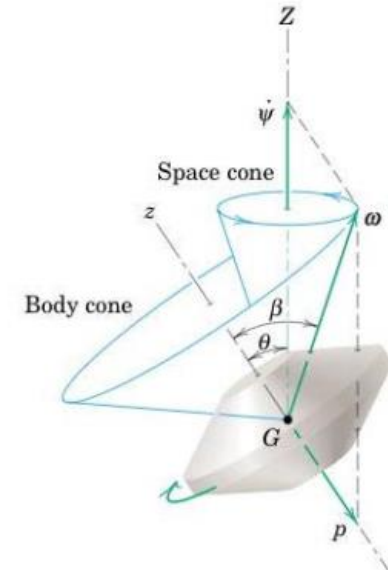


$$\tan \theta = \frac{I_0}{I} \tan \beta$$

$$\dot{\psi} = \frac{Ip}{(I_0 - I) \cos \theta}$$



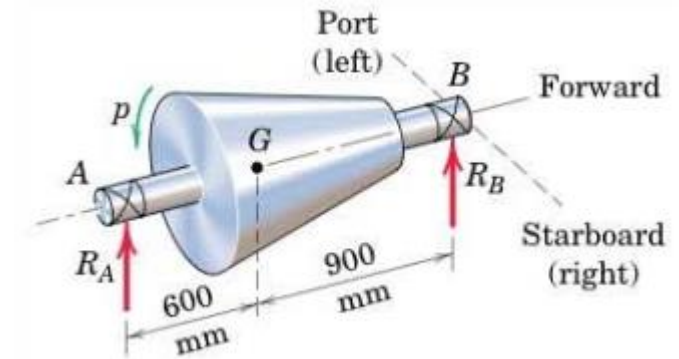
Direct precession $I_0 > I$



Retrograde precession $I_0 < I$

Example

The turbine rotor in a ship's power plant has a mass of 1000 kg, with center of mass at G and a radius of gyration of 200 mm. The rotor shaft is mounted in bearings A and B with its axis in the horizontal fore-and-aft direction and turns counterclockwise at a speed of 500 rev/min when viewed from the stern. Determine the vertical components of the bearing reactions at A and B if the ship is making a turn to port (left) of 400-m radius at a speed of 25 knots (1 knot = 0.514 m/s). Does the bow of the ship tend to rise or fall because of the gyroscopic action?



Example

The student has volunteered to assist in a classroom demonstration involving a momentum wheel which is rapidly spinning with angular speed p as shown. The instructor has asked her to hold the axle of the wheel in the horizontal position shown and then attempt to tilt the axis upward in a vertical plane. What motion tendency of the wheel assembly will the student sense?

